

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The main objective of the Rising Waters project is to design and implement a Machine Learning-driven flood prediction application that evaluates weather related parameters to determine the likelihood of flooding. The system is intended to deliver fast, dependable and accurate flood risk predictions, enabling users and disaster management authorities to make timely decisions and minimize the impact of flood disasters.
Scope	The project is designed to estimate flood risk by utilizing historical weather and rainfall datasets. It covers data collection and preprocessing, training and evaluating multiple machine learning algorithms such as Decision Tree, Random Forest, KNN selecting the best-performing model, integrating it into a Flask-based web application and making the solution accessible through a simple and interactive interface for real-time flood prediction.
Problem Statement	
Description	Flooding is one of the most destructive natural disasters, resulting in loss of life, damage to property and disruption to agriculture and public infrastructure. Existing flood forecasting approaches are often slow, depend on manual analysis or fail to provide accurate early warnings. Therefore, there is a need for a smart and efficient system that can analyze weather and rainfall data to predict flood risk in advance using machine learning techniques.

Impact	The Rising Waters system provides timely and accurate flood risk predictions, allowing residents, farmers, emergency response teams and government authorities to take preventive measures before flooding occurs. By supporting early warning and informed decision-making, the solution helps reduce the impact of floods, enhances disaster preparedness and contributes to the protection of lives, property and critical resources.
Proposed Solution	
Approach	The solution uses a Random Forest Machine Learning algorithm trained on historical flood and weather data. The Flask web application accepts eight meteorological inputs (such as temperature, humidity, cloud cover, and seasonal rainfall totals), standardizes them in real-time using a pre-trained StandardScaler to prevent data leakage, and processes the scaled inputs to compute classification outcomes. Based on the calculated probability percentage, the system dynamically routes the user to either a flood warning or safety page.
Key Features	• Interactive Frontend Layout using HTML, CSS, and Flask featuring an intuitive dashboard. • Machine Learning-Based Flood Prediction • Exploratory Data Analysis (EDA) Dashboard • Relational Database Integration • Automated Cloud Deployment: Custom using PythonAnywhere • Real-time decision making for quicker loan approvals. • Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	GPU is not required; the standard scaler training runs efficiently on standard CPU processing threads)	Standard Dual-Core CPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, joblib (>= 1.2.0), requests

Development Environment	IDE	VS, Code, Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 115 rows $\times$ 11 columns (Approx. 5.5 KB), .csv / .xlsx formats

